Data-Driven Insights into Movie Box Office Success

Team: Group 5
Contributors: Reeves Gonah, Cynthia Jemutai, Eliud Kibet, Stephen Jilani, Mohamed Adan
Project: Phase 2
Date: 23-03-2026
Data Sources: Box Office Mojo, The Numbers, Internet Movie Datase - IMDB, The Movie Database - TMDB and Rotten Tomatoes ___

Introduction

The film industry is a high-risk, high-reward space where production decisions can significantly impact financial outcomes. For a new movie studio entering the market, understanding what drives box office success is critical to making informed investment choices. Rather than relying on intuition or trends alone, data can be used to uncover patterns that consistently influence a film's performance.

This project explores historical movie data to identify the key factors associated with box office success. By combining financial data with attributes such as genre, release timing, and audience reception, the analysis aims to highlight which types of films tend to perform well and why.

The objective is not to build complex predictive models, but to extract clear, interpretable insights that can guide decision-making. The findings from this analysis are used to provide practical recommendations for a new movie studio on what kinds of films to produce, when to release them, and which factors to prioritize in order to maximize revenue potential.

1. Business Understanding

Business Context

- A new movie studio wants to enter the film industry but faces uncertainty around what drives box office success. Using historical movie data can help guide smarter production and release decisions.

Business Objectives

- Identify which types of movies generate the most revenue and the factors influencing movie performance in order to decide on which movies to consider for the pilot project. This yielded the following business specific objectives:
  - To identify genres that are the most popular
  - To establish genre generating most revenue or profit
  - To analyse the effect of budget range on return investment

Business Questions

- The following are the key questions this analysis attempts to answer as they would directly inform procurement decisions:

1. Which movie categories are the most popular? (To identify the market)
2. Which movie categories are the most profitable? (To identify the popular-profitability relationship)
3. What movie budget range maximizes Return On Investment?

Success Criteria

- The business questions above will be considered well answered if three data-backed business recommendations with evident visual support are provided.

2. Data Understanding

Data Source and Description

The data for this project was scraped from multiple movie industry databases and provided to by means of a zipped folder titled ZippedData in this github repository. This included data from Box Office Mojo, The Numbers, IMDb, TMDB, and Rotten Tomatoes as indicated in the project header. Box Office Mojo and The Numbers provide financial data such as box office revenue, while IMDb and TMDB offer descriptive information including genres, release dates, and ratings. Rotten Tomatoes contributes critic and audience scores. Together, these datasets provide a combination of financial performance and movie characteristics, allowing for a more well-rounded analysis of factors associated with box office success.

Initial Data Load

Preliminary analysis on existing data to identify the form and metrics of our data was necessary

```
#Import the necessary Libraries for the project
import pandas as pd
import numpy as np
import sqlite3
import matplotlib.pyplot as plt
import seaborn as sns

# %matplotlib.inline
```

Loading each of the datasets to identify useful data columns

```
#Box office movies as bom
bom = pd.read_csv("bom.movie_gross.csv")
bom.head()
```

```
                                          title studio  domestic_gross
\
0                                Toy Story 3     BV     415000000.0

1                  Alice in Wonderland (2010)     BV     334200000.0
```

```
2  Harry Potter and the Deathly Hallows Part 1     WB     296000000.0

3                                    Inception     WB     292600000.0

4                         Shrek Forever After   P/DW     238700000.0


  foreign_gross  year
0     652000000  2010
1     691300000  2010
2     664300000  2010
3     535700000  2010
4     513900000  2010
```

```
# the movie database as tmdb
tmdb = pd.read_csv("tmdb.movies.csv")
tmdb.head()

##### Observations:
# First column is unnamed
# Release date is in full, we could shorten to the year
```

```
   Unnamed: 0            genre_ids     id original_language  \
0           0      [12, 14, 10751]  12444                en
1           1  [14, 12, 16, 10751]  10191                en
2           2        [12, 28, 878]  10138                en
3           3      [16, 35, 10751]    862                en
4           4        [28, 878, 12]  27205                en

                                original_title  popularity
release_date  \
0  Harry Potter and the Deathly Hallows: Part 1      33.533   2010-11-
19
1                     How to Train Your Dragon      28.734   2010-03-
26
2                                   Iron Man 2      28.515   2010-05-
07
3                                    Toy Story      28.005   1995-11-
22
4                                    Inception      27.920   2010-07-
16

                                         title  vote_average
vote_count
0  Harry Potter and the Deathly Hallows: Part 1           7.7
10788
1                     How to Train Your Dragon           7.7
7610
2                                   Iron Man 2           6.8
12368
```

```
3                                    Toy Story           7.9
10174
4                                    Inception           8.3
22186

#The Numbers as tn
tn = pd.read_csv("tn.movie_budgets.csv")
tn.head()

##### Observations:
#Change 'movie' to title
#Change release date to year
#Potential merge for budget and gross
```

```
   id  release_date                                        movie  \
0   1  Dec 18, 2009                                       Avatar
1   2  May 20, 2011  Pirates of the Caribbean: On Stranger Tides
2   3   Jun 7, 2019                                  Dark Phoenix
3   4   May 1, 2015                  Avengers: Age of Ultron
4   5  Dec 15, 2017           Star Wars Ep. VIII: The Last Jedi

  production_budget domestic_gross worldwide_gross
0      $425,000,000   $760,507,625  $2,776,345,279
1      $410,600,000   $241,063,875  $1,045,663,875
2      $350,000,000    $42,762,350    $149,762,350
3      $330,600,000   $459,005,868  $1,403,013,963
4      $317,000,000   $620,181,382  $1,316,721,747
```

```
# Rotten Tomatoes as rt
rt = pd.read_csv("rt.movie_info.tsv", sep='\t')
rt.head()

##### Observations:
# unlikely to be used
```

```
   id                                           synopsis rating  \
0   1  This gritty, fast-paced, and innovative police...      R
1   3  New York City, not-too-distant-future: Eric Pa...      R
2   5  Illeana Douglas delivers a superb performance ...      R
3   6  Michael Douglas runs afoul of a treacherous su...      R
4   7                                                NaN     NR

                              genre          director  \
0  Action and Adventure|Classics|Drama  William Friedkin
1    Drama|Science Fiction and Fantasy  David Cronenberg
2    Drama|Musical and Performing Arts    Allison Anders
3          Drama|Mystery and Suspense    Barry Levinson
4                      Drama|Romance    Rodney Bennett

                          writer  theater_date      dvd_date
currency  \
```

```
0                   Ernest Tidyman   Oct 9, 1971  Sep 25, 2001
NaN
1     David Cronenberg|Don DeLillo  Aug 17, 2012   Jan 1, 2013
$
2                   Allison Anders  Sep 13, 1996  Apr 18, 2000
NaN
3  Paul Attanasio|Michael Crichton   Dec 9, 1994  Aug 27, 1997
NaN
4                     Giles Cooper           NaN           NaN
NaN

  box_office      runtime             studio
0        NaN  104 minutes                NaN
1    600,000  108 minutes  Entertainment One
2        NaN  116 minutes                NaN
3        NaN  128 minutes                NaN
4        NaN  200 minutes                NaN
```

```
#Rotten Tomato reviews as rt_reviews
# Use encoding to ignore dataset errors and subsequent runtime error
rt_reviews= pd.read_csv('rt.reviews.tsv', sep='\t',encoding='latin1')
rt_reviews.head()
```

```
   id                                             review rating
fresh  \
0   3  A distinctly gallows take on contemporary fina...    3/5
fresh
1   3  It's an allegory in search of a meaning that n...    NaN
rotten
2   3  ... life lived in a bubble in financial dealin...    NaN
fresh
3   3  Continuing along a line introduced in last yea...    NaN
fresh
4   3             ... a perverse twist on neorealism...     NaN
fresh

           critic  top_critic         publisher                date
0      PJ Nabarro           0   Patrick Nabarro  November 10, 2018
1  Annalee Newitz           0           io9.com       May 23, 2018
2    Sean Axmaker           0  Stream on Demand    January 4, 2018
3   Daniel Kasman           0              MUBI  November 16, 2017
4             NaN           0      Cinema Scope   October 12, 2017
```

SQL was necessary to load the final dataset, im.db

```
#Create a connection to the database
conn = sqlite3.connect("im.db")

# Create a cursor to navigate the database
cur = conn.cursor()
```

```
cur.execute("""SELECT name FROM sqlite_master WHERE type =
'table';""")
```

```
<sqlite3.Cursor at 0x2ac10336500>
```

```
#Retrieve the table names
table_names = cur.fetchall()
table_names
```

```
[('movie_basics',),
 ('directors',),
 ('known_for',),
 ('movie_akas',),
 ('movie_ratings',),
 ('persons',),
 ('principals',),
 ('writers',)]
```

```
# Joining the movie ratings table to the movie basics table via SQL
joins
query = """
SELECT *
FROM movie_basics as mb
JOIN movie_ratings as mr
ON mb.movie_id = mr.movie_id
ORDER BY start_year ASC;
"""
#Load the merged data into a new table, Internet Movie Database as
imdb
imdb = pd.read_sql(query, conn)
imdb.head(10)
```

```
    movie_id                            primary_title  \
0  tt0146592                              Pál Adrienn
1  tt0154039                      So Much for Justice!
2  tt0162942             Children of the Green Dragon
3  tt0230212                          The Final Journey
4  tt0312305  Quantum Quest: A Cassini Space Odyssey
5  tt0326592                             The Overnight
6  tt0326965                              In My Sleep
7  tt0331312                      This Wretched Life
8  tt0337882                              Blind Sided
9  tt0393049                        Anderson's Cross

                          original_title  start_year  runtime_minutes
\
0                            Pál Adrienn        2010            136.0

1                        Oda az igazság        2010            100.0

2                A zöld sárkány gyermekei        2010             89.0
```

```
3                       The Final Journey        2010             120.0
4  Quantum Quest: A Cassini Space Odyssey        2010              45.0
5                            The Overnight        2010              88.0
6                              In My Sleep        2010             104.0
7                         This Wretched Life        2010              99.0
8                              Blind Sided        2010               NaN
9                          Anderson's Cross        2010              98.0


                        genres   movie_id  averagerating  numvotes
0                        Drama  tt0146592            6.8       451
1                      History  tt0154039            4.6        64
2                        Drama  tt0162942            6.9       120
3                        Drama  tt0230212            8.8         8
4  Adventure,Animation,Sci-Fi  tt0312305            5.1       287
5                         None  tt0326592            7.5        24
6      Drama,Mystery,Thriller  tt0326965            5.5      1889
7                Comedy,Drama  tt0331312            7.6        59
8           Comedy,Crime,Drama  tt0337882            7.4        14
9         Comedy,Drama,Romance  tt0393049            5.5       106
```

Preliminary Observations and Conclusions

Based on earlier identified business objective and business questions, the following were the columns likely required:

- Genre
- Revenue: domestic gross + worldwide gross
- Budget
- Runtime
- Popularity: rating+ numvotes
- Title
- Year
- Studio

Datasets imdb and tn have all the information/columns necessary. As such, imdb was selected as the primary dataset and tn(the numbers) as the secondary dataset

3.Data Preparation and Cleaning

Data Quality Assessment and Data Selection

The selected datasets were reloaded and examined to assess the integrity and quality of data

```
# Reloading the selected dataset: imdb
imdb
```

```
        movie_id                              primary_title  \
0      tt0146592                                Pál Adrienn
1      tt0154039                        So Much for Justice!
2      tt0162942               Children of the Green Dragon
3      tt0230212                           The Final Journey
4      tt0312305  Quantum Quest: A Cassini Space Odyssey
...          ...                                        ...
73851  tt9913056                               Swarm Season
73852  tt9913084                             Diabolik sono io
73853  tt9914286                           Sokagin Çocuklari
73854  tt9914942                 La vida sense la Sara Amat
73855  tt9916160                                 Drømmeland

                                original_title  start_year
runtime_minutes  \
0                                   Pál Adrienn        2010
136.0
1                                Oda az igazság        2010
100.0
2                     A zöld sárkány gyermekei        2010
89.0
3                              The Final Journey        2010
120.0
4      Quantum Quest: A Cassini Space Odyssey        2010
45.0
...                                          ...         ...
...
73851                              Swarm Season        2019
86.0
73852                            Diabolik sono io        2019
75.0
73853                           Sokagin Çocuklari        2019
98.0
73854                 La vida sense la Sara Amat        2019
NaN
73855                               Drømmeland        2019
72.0

                            genres   movie_id  averagerating  numvotes

0                            Drama  tt0146592            6.8       451
```

```
1                          History  tt0154039            4.6         64
2                            Drama  tt0162942            6.9        120
3                            Drama  tt0230212            8.8          8
4      Adventure,Animation,Sci-Fi  tt0312305            5.1        287
...                             ...        ...            ...        ...
73851                  Documentary  tt9913056            6.2          5
73852                  Documentary  tt9913084            6.2          6
73853                Drama,Family  tt9914286            8.7        136
73854                         None  tt9914942            6.6          5
73855                  Documentary  tt9916160            6.5         11

[73856 rows x 9 columns]
```

```
# missing values appear on runtime column, investigate

imdb['runtime_minutes'].isna().sum()

###Observation:
# Many missing values from runtime column (roughly 10%)
# proceed for now
```

```
7620
```

The dataset was then loaded under the name primary_table, with only columns deemed necessary for analysis

```
# Listing necessary columns
columns = ['original_title',
'start_year','runtime_minutes','genres','averagerating','numvotes']
#Assigning new dataframe
primary_table = imdb[columns]
#primary_table
```

And the column titles were renamed

```
#Renaming the column titles
new_titles =
['Movie_Title','Year','Runtime','Genres','Average_Rating','Number_of_V
```

```
otes']
primary_table.columns = new_titles
```

The same was done for the tn(the numbers) dataset

```
# Reloading the selected dataset: tn
tn
```

```
      id  release_date                                          movie  \
0      1  Dec 18, 2009                                         Avatar
1      2  May 20, 2011  Pirates of the Caribbean: On Stranger Tides
2      3   Jun 7, 2019                                   Dark Phoenix
3      4   May 1, 2015                    Avengers: Age of Ultron
4      5  Dec 15, 2017          Star Wars Ep. VIII: The Last Jedi
...   ..           ...                                          ...
5777  78  Dec 31, 2018                                      Red 11
5778  79   Apr 2, 1999                                   Following
5779  80  Jul 13, 2005               Return to the Land of Wonders
5780  81  Sep 29, 2015                        A Plague So Pleasant
5781  82   Aug 5, 2005                           My Date With Drew

     production_budget domestic_gross worldwide_gross
0         $425,000,000   $760,507,625  $2,776,345,279
1         $410,600,000   $241,063,875  $1,045,663,875
2         $350,000,000    $42,762,350    $149,762,350
3         $330,600,000   $459,005,868  $1,403,013,963
4         $317,000,000   $620,181,382  $1,316,721,747
...                ...            ...             ...
5777             $7,000            $0              $0
5778             $6,000        $48,482        $240,495
5779             $5,000         $1,338          $1,338
5780             $1,400            $0              $0
5781             $1,100       $181,041        $181,041

[5782 rows x 6 columns]
```

```
#Creating our secondary dataframe using only the columns we require
from the tn database
columns_1 = ['movie','production_budget',
'domestic_gross','worldwide_gross']

secondary_table = tn[columns_1]
#secondary_table

#Renaming the column titles
new_titles_1 =
['Movie_Title','Production_Budget','Domestic_Revenue','Worldwide_Reven
ue']
secondary_table.columns = new_titles_1
secondary_table.head()
```

```
                                    Movie_Title Production_Budget  \
0                                        Avatar      $425,000,000
1  Pirates of the Caribbean: On Stranger Tides      $410,600,000
2                                  Dark Phoenix      $350,000,000
3                       Avengers: Age of Ultron      $330,600,000
4            Star Wars Ep. VIII: The Last Jedi      $317,000,000

  Domestic_Revenue Worldwide_Revenue
0     $760,507,625    $2,776,345,279
1     $241,063,875    $1,045,663,875
2      $42,762,350      $149,762,350
3     $459,005,868    $1,403,013,963
4     $620,181,382    $1,316,721,747
```

Data Merging

Before cleaning all rows and columns, merging was necessary so as only to be left with rows of movies appearing on both datasets. The movie title was selected as the merging key and as such needed to be cleaned on both tables prior to merging

```
#Create a function to remove whitespace from movie titles
def remove_whitespace(column):
    column = column.str.strip()
    return column

# Cleaning titles from the primary table

primary_table = primary_table.copy() # to avoid warning

#Remove whitespace from title column
primary_table['Movie_Title'] =
remove_whitespace(primary_table['Movie_Title'])

# Cleaning titles from the primary table
secondary_table = secondary_table.copy()

#Remove whitespace from title column
secondary_table['Movie_Title'] =
remove_whitespace(secondary_table['Movie_Title'])
```

With clean movie titles, merging was done

```
# Using the Movie Title column, lets merge the two tables into one
movie database
movie_database = pd.merge(primary_table,secondary_table)
movie_database

               Movie_Title  Year  Runtime                        Genres
\
0            The Overnight  2010     88.0                          None
```

```
1            The Overnight  2015     79.0              Comedy,Mystery
2         Anderson's Cross  2010     98.0        Comedy,Drama,Romance
3                  Tangled  2010    100.0  Adventure,Animation,Comedy
4      Dinner for Schmucks  2010    114.0                      Comedy
...                    ...   ...      ...                         ...
2633    Happy Death Day 2U  2019    100.0        Drama,Horror,Mystery
2634  Blinded by the Light  2019    117.0      Biography,Comedy,Drama
2635                Heroes  2019     88.0                 Documentary
2636                  Push  2019     92.0                 Documentary
2637             Unplanned  2019    106.0             Biography,Drama


      Average_Rating  Number_of_Votes Production_Budget
Domestic_Revenue  \
0                7.5               24          $200,000
$1,109,808
1                6.1            14828          $200,000
$1,109,808
2                5.5              106          $300,000
$0
3                7.8           366366      $260,000,000
$200,821,936
4                5.9            91546       $69,000,000
$73,026,337
...              ...              ...               ...               .
..
2633             6.3            27462        $9,000,000
$28,051,045
2634             6.2              173       $15,000,000
$0
2635             7.3                7          $400,000
$655,538
2636             7.3               33       $38,000,000
$31,811,527
2637             6.3             5945        $6,000,000
$18,107,621

     Worldwide_Revenue
0           $1,165,996
1           $1,165,996
```

```
2                    $0
3          $586,477,240
4           $86,796,502
...                 ...
2633        $64,179,495
2634                 $0
2635            $655,538
2636        $49,678,401
2637        $18,107,621

[2638 rows x 9 columns]
```

This new table was renamed as movie database. Investigating the columns:

```
movie_database.isna().sum()
```

```
Movie_Title            0
Year                   0
Runtime              106
Genres                 5
Average_Rating         0
Number_of_Votes        0
Production_Budget      0
Domestic_Revenue       0
Worldwide_Revenue      0
dtype: int64
```

```
movie_database['Runtime'].describe()
```

```
count    2532.000000
mean      103.049368
std        20.360082
min         3.000000
25%        90.000000
50%       101.000000
75%       114.000000
max       189.000000
Name: Runtime, dtype: float64
```

```
#Filter those with missing genres
missing_genres = movie_database['Genres'].isna()
movie_database.loc[missing_genres]
```

```
                Movie_Title  Year  Runtime Genres  Average_Rating  \
0             The Overnight  2010     88.0   None             7.5
55               Robin Hood  2018      NaN   None             7.6
56               Robin Hood  2018      NaN   None             7.6
85        The Bounty Hunter  2010      NaN   None             6.3
2102  When the Bough Breaks  2017     93.0   None             6.1
```

```
      Number_of_Votes Production_Budget Domestic_Revenue
Worldwide_Revenue
0                  24          $200,000       $1,109,808
$1,165,996
55                  5      $210,000,000     $105,487,148
$322,459,006
56                  5       $99,000,000      $30,824,628
$84,747,441
85                 29       $45,000,000      $67,061,228
$135,808,837
2102                8       $10,000,000      $29,747,603
$30,768,449
```

Data Cleaning

A few issues were noted with the new dataset, such as column formatting and missing values

```
# Further Data Cleaning
# Checking the datatypes
movie_database.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 2638 entries, 0 to 2637
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Movie_Title        2638 non-null   object
 1   Year               2638 non-null   int64
 2   Runtime            2532 non-null   float64
 3   Genres             2633 non-null   object
 4   Average_Rating     2638 non-null   float64
 5   Number_of_Votes    2638 non-null   int64
 6   Production_Budget  2638 non-null   object
 7   Domestic_Revenue   2638 non-null   object
 8   Worldwide_Revenue  2638 non-null   object
dtypes: float64(2), int64(2), object(5)
memory usage: 206.1+ KB

# Convert Revenue and Budget columns to integer

# Remove $ sign first
movie_database['Production_Budget'] =
movie_database['Production_Budget'].str.replace(r'[$,]
+','',regex=True)
movie_database['Domestic_Revenue'] =
movie_database['Domestic_Revenue'].str.replace(r'[$,]+','',regex=True)
movie_database['Worldwide_Revenue'] =
movie_database['Worldwide_Revenue'].str.replace(r'[$,]
+','',regex=True)
```

```
# Convert to integer
movie_database['Production_Budget'] =
movie_database['Production_Budget'].astype(int)
movie_database['Domestic_Revenue'] =
movie_database['Domestic_Revenue'].astype(int)
movie_database['Worldwide_Revenue'] =
pd.to_numeric(movie_database['Worldwide_Revenue'],errors='coerce')

# Confirm changes
movie_database.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2638 entries, 0 to 2637
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Movie_Title        2638 non-null   object
 1   Year               2638 non-null   int64
 2   Runtime            2532 non-null   float64
 3   Genres             2633 non-null   object
 4   Average_Rating     2638 non-null   float64
 5   Number_of_Votes    2638 non-null   int64
 6   Production_Budget  2638 non-null   int32
 7   Domestic_Revenue   2638 non-null   int32
 8   Worldwide_Revenue  2638 non-null   int64
dtypes: float64(2), int32(2), int64(3), object(2)
memory usage: 185.5+ KB
```

Additional Columns Creation

An additional 'Profit' column was deemed necessary. This was done by deducting the movie budget from its total revenue

```
#Creating a new calculated column for profit
movie_database['Profit'] = movie_database['Worldwide_Revenue'] -
movie_database['Production_Budget']
movie_database.head()
```

```
           Movie_Title  Year  Runtime                        Genres  \
0        The Overnight  2010     88.0                          None
1        The Overnight  2015     79.0                Comedy,Mystery
2     Anderson's Cross  2010     98.0         Comedy,Drama,Romance
3              Tangled  2010    100.0  Adventure,Animation,Comedy
4  Dinner for Schmucks  2010    114.0                        Comedy

   Average_Rating  Number_of_Votes  Production_Budget
Domestic_Revenue  \
0             7.5               24             200000
1109808
```

```
1              6.1             14828              200000
1109808
2              5.5               106              300000
0
3              7.8            366366           260000000
200821936
4              5.9             91546            69000000
73026337

   Worldwide_Revenue     Profit          ROI
0            1165996     965996  1037.902000
1            1165996     965996  1037.902000
2                  0    -300000  -100.000000
3          586477240  326477240   202.807375
4           86796502   17796502   131.627303
```

Simiilarly, a return on investment column was deemed as important, as it is more indicative on movie profitability as not all movies have similar production budgets

```
movie_database['ROI'] =
(movie_database['Profit']/movie_database['Production_Budget'])
movie_database.head()

           Movie_Title  Year  Runtime                        Genres  \
0        The Overnight  2010     88.0                          None
1        The Overnight  2015     79.0               Comedy,Mystery
2    Anderson's Cross  2010     98.0         Comedy,Drama,Romance
3             Tangled  2010    100.0  Adventure,Animation,Comedy
4  Dinner for Schmucks  2010    114.0                        Comedy

   Average_Rating  Number_of_Votes  Production_Budget
Domestic_Revenue  \
0             7.5               24             200000
1109808
1             6.1            14828             200000
1109808
2             5.5              106             300000
0
3             7.8           366366          260000000
200821936
4             5.9            91546           69000000
73026337

   Worldwide_Revenue     Profit       ROI
0            1165996     965996  4.829980
1            1165996     965996  4.829980
2                  0    -300000 -1.000000
3          586477240  326477240  1.255682
4           86796502   17796502  0.257920
```

Final EDA Remarks

- The runtime column now has only 106 null/missing values. These rows were not dropped as all other columns had their data present, and null values will not be used in computing descriptive statistics about movie runtime such as mean/count
- There are some movie titles repeating. Since the year nor rating is the same, it was assumed that movies are different and as such those rows are not dropped
- Five movies are missing genres. All other columns for these rows contain data. They are not dropped as a result

4.Analysis

Based on the earlier identified business problems, Further analysis needs to be done to identify the relationships between data.

1.) Business Objective I: To identify genres that are the most popular

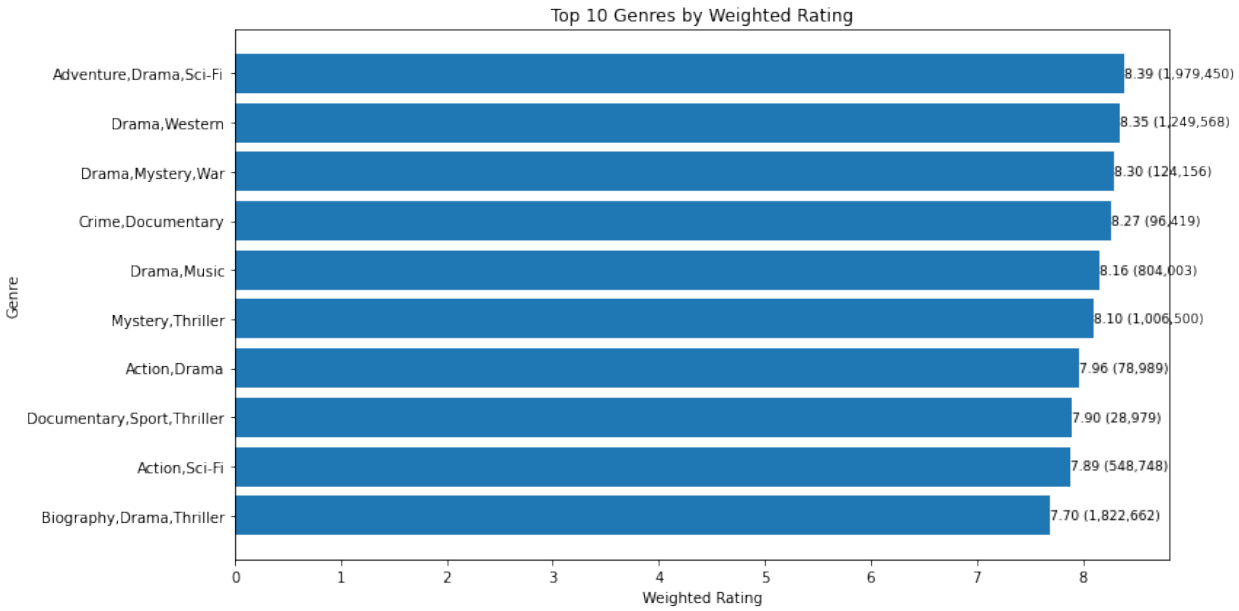
Establish the top movie genres according to voters, based on rating.

The purpose of this is to identify where the market is/ which movies appeal to the larger audience, by means of movie quality (rating), genre and popularity(number of voters, a high number indicates it was also widely watched)

- Movies are grouped by genre, treating a combination of genres as a genre itself.
- Because not all movies/ movie genres have the same number of votes, the **weighted average** is computed to allow us to fairly determine the most popular genre of movies
- Movies with a vote count of less than 5 thousand are considered to not have enough votes to be considered fair to deduce judgement from

```
# Group the movies by genre
# Weighted Average Formula = sum(Average Rating * Number of Votes)/
sum(Total Votes for genre group)
grouped = (
    movie_database
    .groupby("Genres")
    .apply(lambda x: (x["Average_Rating"] *
x["Number_of_Votes"]).sum() / x["Number_of_Votes"].sum())
    .to_frame(name="weighted_rating")
)

# Keep total voters for context
grouped["total_voters"] = movie_database.groupby("Genres")
["Number_of_Votes"].sum()
```

```
# Keep the genre categories where the total voters are less than 5000
grouped = grouped[grouped["total_voters"] > 5000]

# Retrieve the top 10 values and store them in a database called top
10 (using descending order)
top_10 = grouped.sort_values(by="weighted_rating",
ascending=False).head(10)

# Sort for horizontal plotting (for the visualization)
top_10 = top_10.sort_values(by="weighted_rating")

# Create the plot
fig, ax = plt.subplots(figsize=(12, 6))

# barh for horizontal bar graph
ax.barh(top_10.index, top_10["weighted_rating"])

# Set title and labels
ax.set_title("Top 10 Genres by Weighted Rating")
ax.set_xlabel("Weighted Rating")
ax.set_ylabel("Genre")

# Add labels (rating + voters) (found from online forum, adapted for
this case)
for i, (rating, voters) in enumerate(zip(top_10["weighted_rating"],
top_10["total_voters"])):
    ax.text(rating, i, f"{rating:.2f} ({voters:,})", va="center",
fontsize=9)

plt.tight_layout()
plt.show()
```



From the Graph above, we can conclude that the most popular movies are of the drama genre in combination with high-engagement elements like sci-fi, adventure, or mystery. These have very high vote counts indicating a larger audience, and high ratings (above 8.0) which show a prime market for movie creation. The key observations from this analyis and its resulting visualization can be summarized as follows;

Key Observations:

- Adventure, Drama, Sci-Fi lead in both rating and total voters.
- Genres like Drama, Western and Mystery, Thriller also achieve strong ratings at scale, indicating broad audience appeal.
- In contrast, some high-rated genres with low voter counts suggest niche interest (e.g War, Sport, Documentary)

2.) Business Objective II: To establish genre generating most revenue or profit

Establish the relationship between the movie genre and the total profit to identify those with the most upside for the new in-house movie studio.
Focus is done on the top 10 movie genres identified earlier

```
# Creating a dataframe to group the movie by genres and compute the average roi
genre_profitability = (
    movie_database.groupby('Genres')['ROI']
    .mean()
    .sort_values(ascending=False)
    .reset_index()
)
genre_profitability.head()
```

```
                   Genres        ROI
0     Crime,Drama,Family  62.119120
1  Biography,Documentary  52.616649
2   Drama,Family,Fantasy  47.260224
3    Action,Comedy,Drama  32.876039
4                  Crime  27.873195
```

```
# Recall the earlier created genre popularity table
grouped.head()
```

```
                            weighted_rating  total_voters
Genres
Action                             5.434337         60425
Action,Adventure                   5.800000          6955
Action,Adventure,Animation         7.604005       3046710
Action,Adventure,Biography         7.650532        945069
Action,Adventure,Comedy            7.312251       6051788
```

```
# Merging the popularity table and the average roi table for
comparison purposes
#use join
genre_comparison = pd.merge(
    grouped,
    genre_profitability,
    on='Genres',
    how='inner'
)
genre_comparison.head()
```

```
                      Genres  weighted_rating  total_voters       ROI
0                     Action         5.434337         60425  2.175336
1           Action,Adventure         5.800000          6955 -0.997384
2  Action,Adventure,Animation         7.604005       3046710  2.454735
3  Action,Adventure,Biography         7.650532        945069  1.475850
4     Action,Adventure,Comedy         7.312251       6051788  3.310098
```

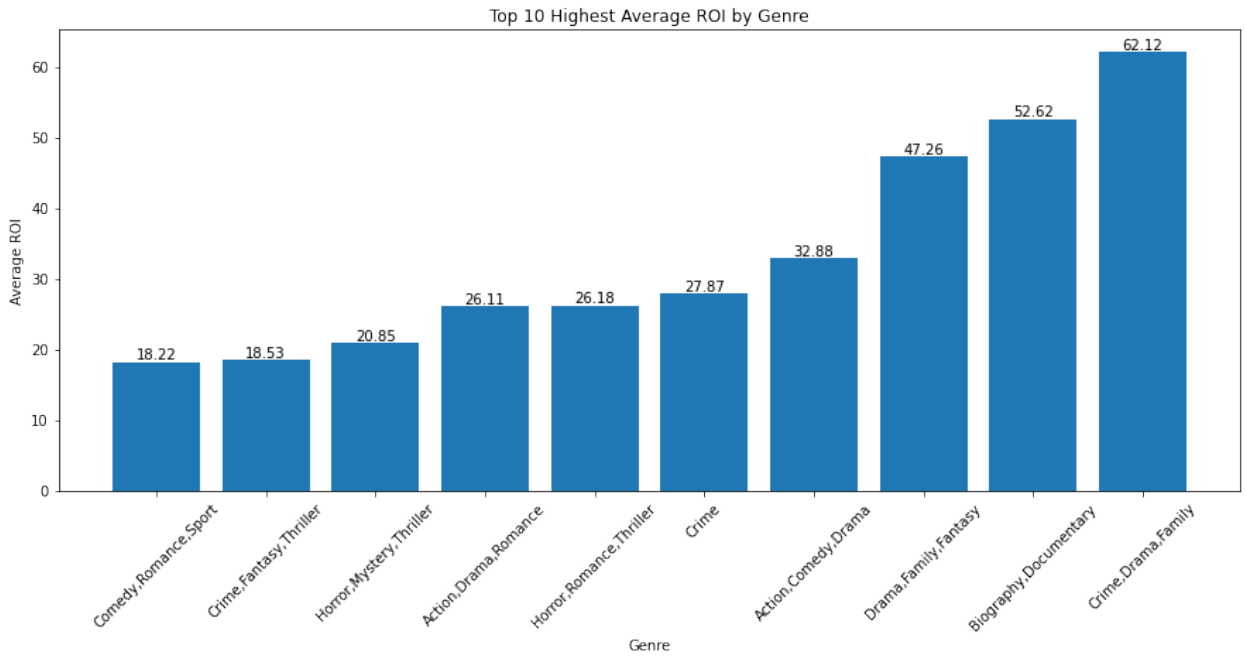
```
# Now sorting them in descending order
genre_comparison['popularity_rank'] =
genre_comparison['weighted_rating'].rank(ascending=False,
method='min')
genre_comparison['profitability_rank'] =
genre_comparison['ROI'].rank(ascending=False, method='min')

genre_comparison = genre_comparison.sort_values('popularity_rank')
genre_comparison.head(10)
```

```
                        Genres  weighted_rating  total_voters
ROI  \
80       Adventure,Drama,Sci-Fi         8.393847       1979450
4.052996
200               Drama,Western         8.345026       1249568
```

```
0.982682
187           Drama,Mystery,War          8.300000         124156
1.358580
147           Crime,Documentary          8.267729          96419
1.563472
180                 Drama,Music          8.155639         804003
1.550099
214            Mystery,Thriller          8.098578        1006500
0.249034
37                 Action,Drama          7.961615          78989
1.201406
161  Documentary,Sport,Thriller          7.900000          28979 -
1.000000
56                Action,Sci-Fi          7.889321         548748
0.303344
107    Biography,Drama,Thriller          7.696396        1822662
2.189535

     popularity_rank  profitability_rank
80               1.0                35.0
200              2.0               133.0
187              3.0               116.0
147              4.0               104.0
180              5.0               105.0
214              6.0               170.0
37               7.0               123.0
161              8.0               217.0
56               9.0               168.0
107             10.0                74.0
```

```
# Visualizing the results
#Start by visualizing the movie genres with the highest return on
investment

# Create the plot figure
plt.figure(figsize=(15, 6))

#Get the movie genres with the highest ROI (Top 10)
top_10_ROI = genre_profitability.sort_values('ROI',
ascending=False).head(10)

# Sort for horizontal plotting (for the visualization)
top_10_ROI = top_10_ROI.sort_values(by="ROI")

bars = plt.bar(top_10_ROI['Genres'], top_10_ROI['ROI'])
# Add value labels on top of each bar
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2,
```

```
        height,
        f'{height:.2f}',
        ha='center',
        va='bottom'
    )

plt.ylabel('Average ROI')
plt.xlabel('Genre')
plt.xticks(rotation=45)
plt.title('Top 10 Highest Average ROI by Genre')
plt.show()
```



Key Observations

- The top ROI genres are spread across a mix of genres
- Many of the highest-performing films are multi-genre combinations
- Popularity does not necessarily translate into strong profit performance.

3.) Business Objective II: To analyse the effect of budget range on return investment

Identify the budget range that provides the most upsside for the new movie studio, to prevent or minimize sunken costs.

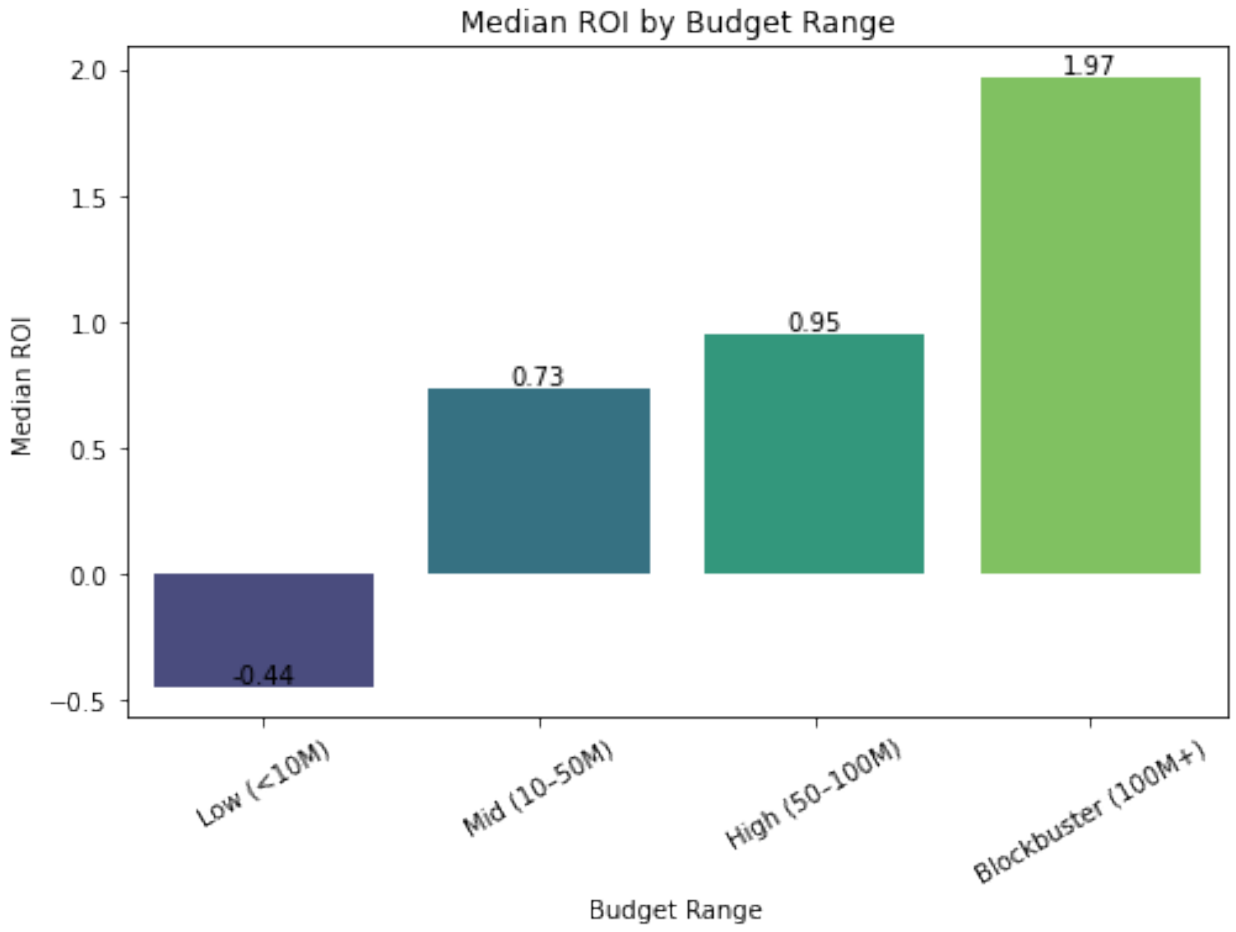
```
# Create Budget bins to group movies by
bins = [0, 10e6, 50e6, 100e6, np.inf]
```

```
labels = ['Low (<10M)', 'Mid (10–50M)', 'High (50–100M)', 'Blockbuster
(100M+)']

movie_database['budget_range'] =
pd.cut(movie_database['Production_Budget'], bins=bins, labels=labels)

# Then Aggregate metrics (mean, median, count) by budget range
summary = movie_database.groupby('budget_range').agg({
    'ROI': ['mean', 'median'],
    'Profit': ['mean', 'median'],
    'Production_Budget': 'count'
}).reset_index()

summary.columns = ['budget_range', 'roi_mean', 'roi_median',
'profit_mean', 'profit_median', 'count']

summary.head()
```

```
          budget_range  roi_mean  roi_median   profit_mean
profit_median  \
0           Low (<10M)  5.267657   -0.444355  1.296276e+07         -
172398
1         Mid (10–50M)  1.722058    0.732457  4.193168e+07
17811261
2       High (50–100M)  1.685814    0.953429  1.228031e+08
71487563
3  Blockbuster (100M+)  2.286905    1.969215  3.754361e+08
300391735

   count
0   1055
1   1056
2    293
3    234
```

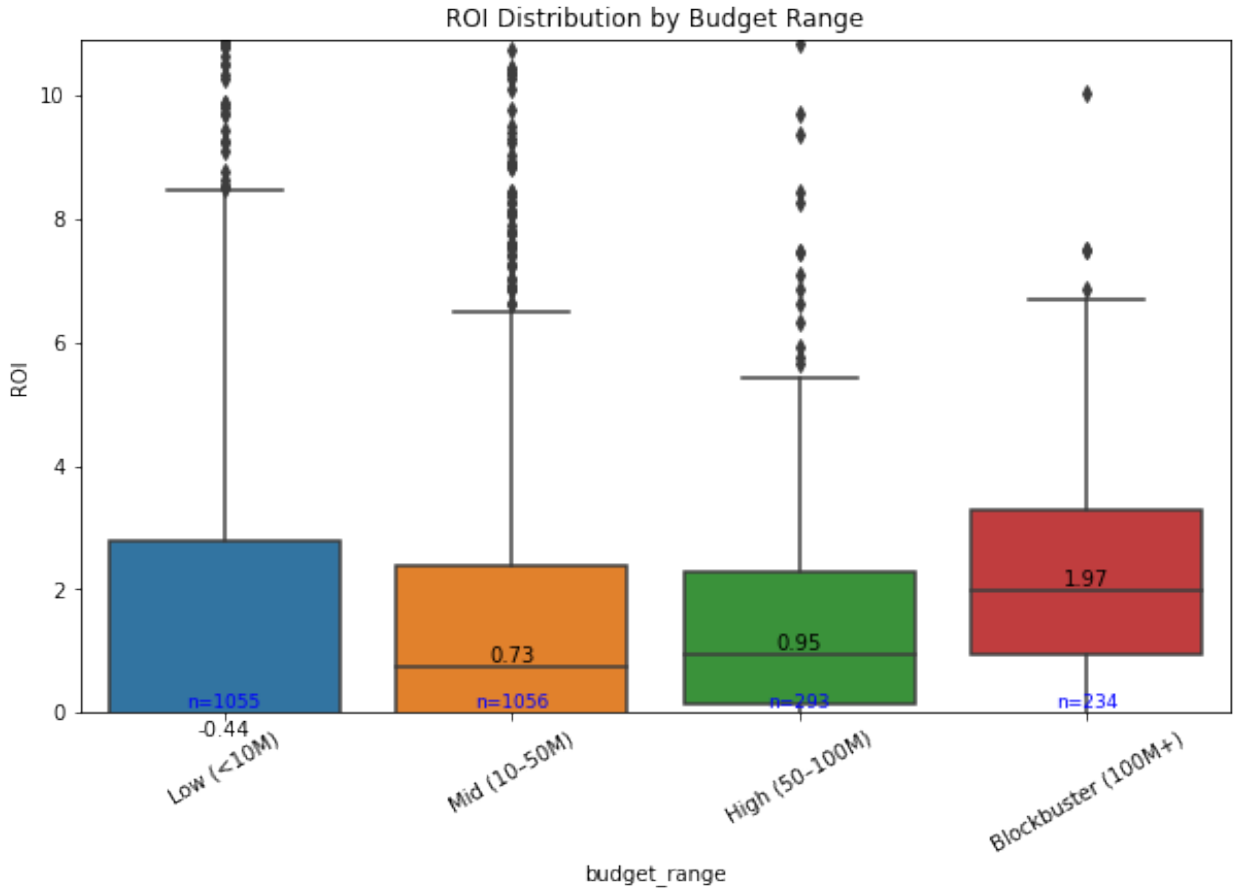
```
# Option 1: Bar plot
# Create plot figure
plt.figure(figsize=(8,5))

# select data into bar graph (median ROI is selected to eliminate bias
due to outliers)
ax = sns.barplot(data=summary, x='budget_range', y='roi_median',
palette='viridis')

# Add labels on top of each bar
for p in ax.patches:
    height = p.get_height()
    ax.text(
        p.get_x() + p.get_width() / 2,  # x position
        height,                         # y position
        f'{height:.2f}',               # label (2 decimal places)
```

```
        ha='center',
        va='bottom'
    )

# Plot Labels
plt.title('Median ROI by Budget Range')
plt.ylabel('Median ROI')
plt.xlabel('Budget Range')
plt.xticks(rotation=30)
plt.show()
```




```
# Option 2: Box plots for distribution illustration
plt.figure(figsize=(10,6))
ax = sns.boxplot(data=movie_database, x='budget_range', y='ROI')

# Add median labels
medians = movie_database.groupby('budget_range')['ROI'].median()

for i, median in enumerate(medians):
    ax.text(i, median, f'{median:.2f}',
            ha='center', va='bottom', color='black', fontsize=10)
```

```
counts = movie_database['budget_range'].value_counts().sort_index()

for i, count in enumerate(counts):
    ax.text(i, 0, f'n={count}',
            ha='center', va='bottom', fontsize=9, color='blue')

plt.title('ROI Distribution by Budget Range')
plt.ylim(0, movie_database['ROI'].quantile(0.95))
plt.xticks(rotation=30)

plt.show()
```

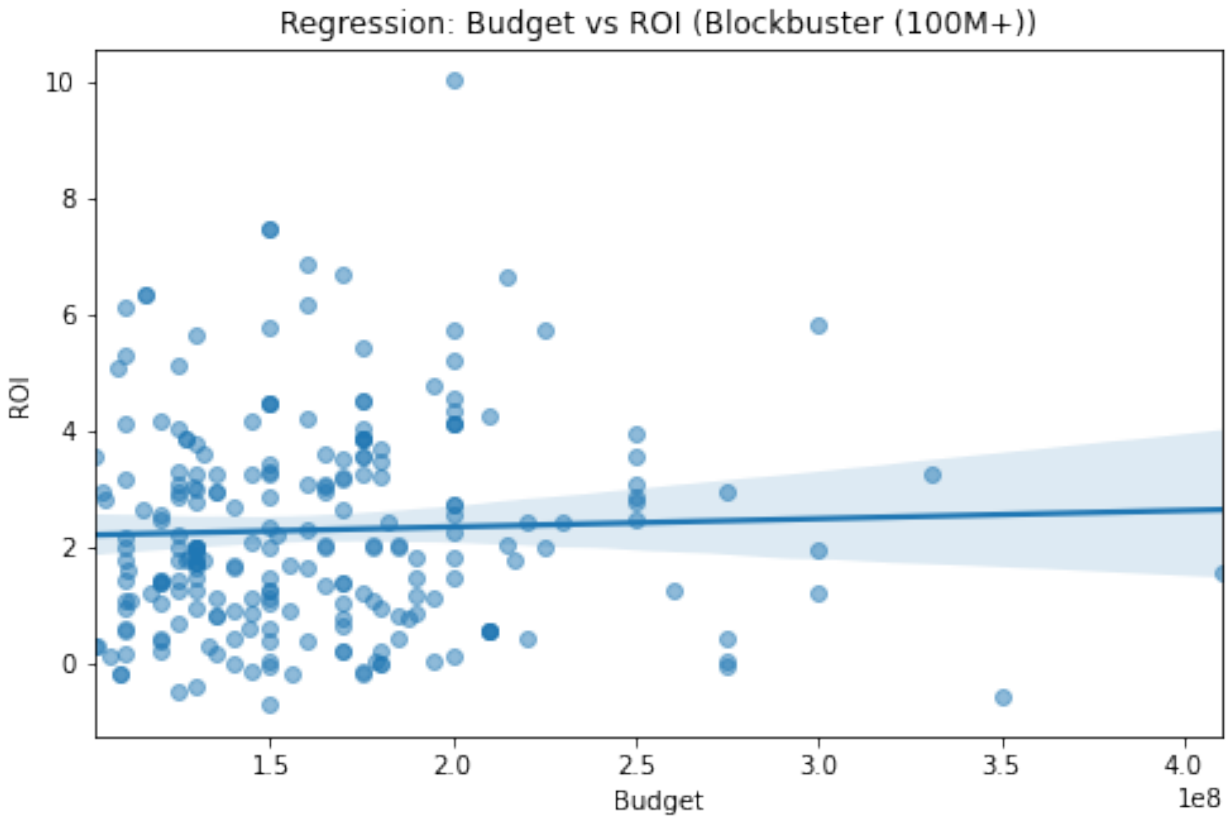



```
# Identify the best budget range
best_range = summary.loc[summary['roi_median'].idxmax(),
'budget_range']
print("Best budget range based on median ROI:", best_range)
```

```
Best budget range based on median ROI: Blockbuster (100M+)
```

```
#Filter the data for that budget range
best_df = movie_database[movie_database['budget_range'] == best_range]
```

```
# Create a regression plot for the selected budget range
plt.figure(figsize=(8,5))
sns.regplot(data=best_df, x='Production_Budget', y='ROI',
scatter_kws={'alpha':0.5})
plt.title(f'Regression: Budget vs ROI ({best_range})')
plt.xlabel('Budget')
plt.ylabel('ROI')
plt.show()
```



Assesment of the Return on investment was based on the following scale:

- Low budget: < 10M
- Medium: 10M–50M
- High: 50M–100M
- Blockbuster: 100M+

Key Observations

- ROI increases with budget, with blockbuster film range (100+ M) delivering the highest median returns.
- Low-budget films (<10 M) consistently generate the lowest ROI, suggesting limited revenue scalability.

- `Blockbuster films (100+ M) are few, indicating high selectivity and potential barriers to entry.`

5.Business Recommendations

Recommendations were done based on the specific objectives i.e the business questions identified earlier.

1.Identifying genres that are the most popular

- Focus on drama-driven hybrid genres with mass appeal (such as a combination with adventure, sci-fi)is likely to yield the best returns, while niche genres (such as documentary, sports, thriller) are better suited for targeted audiences.

2.To establish the genre that generates the most revenue/Profit

- The studio should not rely on popularity alone when choosing what films to produce. Instead, it should focus more on genres and genre combinations (especially drama with other genres such as action comedy and crime) which have strong ROI potential

3.To analyse the effect of budget range on return on investment (ROI)

- Focus on selectively investing in high-budget (100+ M) films, prioritizing projects with strong success indicators (e.g., established franchises or proven talent), to maximize ROI while managing the risks associated with large-scale production.

6. Conclusion and Next Steps

Overall, the analysis shows that successful film production decisions require a balance between targetting audience appetite, revenue potential, and strategic investment. While popular genres attract wide audiences, the most profitable outcomes come from thoughtful genre combinations and well-targeted high-budget projects. In particular, blending drama genre with commercially strong genres (such as action and adventure) and selectively backing high-budget productions offers the best opportunity to maximize returns.

Given time and extra resources, the studio should deepen this analysis by exploring additional drivers of success, such as cast, directors, release timing, and franchise effects, to better inform high-budget investment decisions. Incorporating predictive modeling and continuously updating insights with new data will also help refine strategy and ensure more consistent, data-driven film production choices.